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Falcon 9 CRS-10 Vehicle Simulator and Navigation

Topics in Aerospace Engineering - Team 5

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1 Introduction and Project Scope

1.1 Project 2 objectives

Project 2 develops a physics-based Falcon 9 simulator and navigation driven by simulated sensor measurements. Forces, moments, changing mass properties and numerical integration generate motion; imperfect observations of that motion drive state estimation. The course defines this environment as the computational foundation for guidance in Project 3 and integrated control in Project 4 [11, slides 2–4].

The required models cover equations of motion, coordinate systems, gravity, atmosphere, propulsion, aerodynamics, mass/inertia variation and configuration changes. Navigation addresses position, velocity, attitude and angular velocity. Evaluation therefore separates model verification, comparison with public flight evidence, and estimation error relative to simulation truth [11, slides 4–6].

1.2 Why CRS-10?

CRS-10 is SpaceX's tenth contracted Dragon cargo-resupply mission to the ISS. Official material identifies LC-39A launch, first-stage recovery at LZ-1, and approximate cutoff, separation and restart times [1]. Its webcast and published ascent digitization provide a quantitative reference [13, 14]. Common-stack ascent, booster return and upper-stage continuation exercise both continuous propagation and configuration changes.

The modeled payload is one loaded Dragon/trunk assembly, not a satellite and separate fairing. NASA's 2490 kg cargo inventory is included within that assembly, not its total mass [2]. Figure 1 identifies the hardware and official mission document; Section 3 qualifies the mixed-source internal properties.



Figure 1 CRS-10 reference configuration. Left: Falcon 9 and Dragon at LC-39A before launch, NASA/Glenn Benson [12]. Right: header and mission-patch excerpts from the official SpaceX press kit [1]. The images identify the hardware and source; they do not determine the internal mass distribution.

1.3 Demonstration and validation scope

Both children inherit the propagated separation state. The booster returns to geometric contact, while the upper-stage/Dragon child continues under assumed commands. First-stage navigation uses gyro-propagated attitude and GNSS/LiDAR updates with a gated star-tracker interface. Truth supplies virtual observations, assumed initial alignment and evaluation, rather than continuous estimator attitude input.

External comparison covers approximately 0–140 s of E-class digitized ascent readouts. The ascent operating point and pitch timescale were tuned against this same record; their agreement is calibration evidence, not independent validation. The later chapters retain that distinction when reporting flight, contact and navigation results. Return robustness, reconstructed SpaceX guidance and validated orbital insertion are outside the demonstrated scope.

2 Simulation Architecture and Shared Conventions

2.1 Physical truth, measurements, and estimates

The physical branch computes loads and mass properties and integrates vehicle motion. The measurement branch samples that truth through sensor error and availability models. Navigation propagates its own attitude and estimates the IMU-station motion. Comparison uses the same station and time. Figure 2 separates these interfaces.

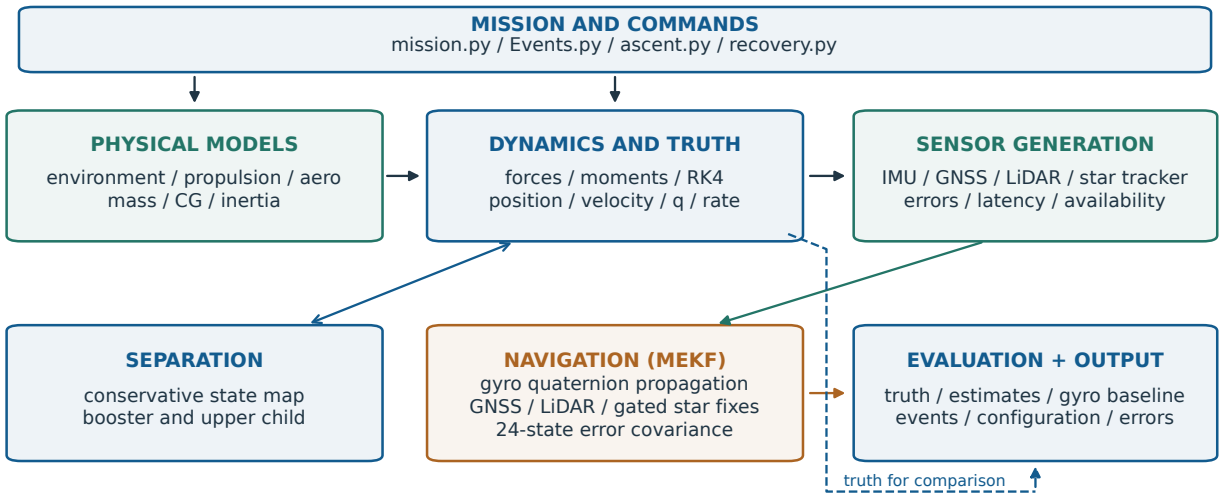


Figure 2 Implemented data flow. The MEKF uses simulated sensor measurements and its own propagated attitude. Truth supplies sensor generation and evaluation; initial alignment is assumed. Return commands use physical-state feedback, while navigation is evaluated separately.

The mission layer selects the initial configuration, commands, and accepted phase. A throttle command becomes thrust and flow through propulsion; an aerodynamic configuration and fin command become force and moment through aerodynamics. The dynamics model then calculates the resulting motion. Neither the force provider nor the mass model prescribes the reference altitude/speed history. Table 1 assigns the corresponding implementation responsibilities.

Function	Main modules	Interface

Vehicle properties	`VehicleData.py`
	`vehicle_mass_model.py`
	`mass_properties.py`
		Trial inventory to mass, CG, inertia, and derivatives.

Environment and loads	`Environement.py`
	`propulsion.py`
	`Aerodynamics.py`
	`load_interface.py`
		Trial state/commands to loads and exhaust data.

Motion and configuration	`dynamics.py`
	`frames.py`
	`staging.py`
		Derivative, integrated state, and inherited child states.

Measurement and estimation	`sensors.py` `kalman.py`	Sensor-station truth to observations and separate estimates.

Mission execution	`main.py`
	`mission.py`
	`Events.py`
		Commands, accepted clocks/events, state synchronization, and saved output.

Table 1 Responsibilities of the shared implementation. Configuration and scenario decisions are kept separate from reusable component arithmetic and equations of motion.

2.2 Frames, origins, and attitude convention

Frame B is right-handed and fixed to the vehicle, with $+x_B$ toward the nose. Frame E is Earth-centred, launch-aligned, and rotating with Earth. Code identifiers call it ECEF, but its axes are not conventional Greenwich-axis

ECEF. At the launch site, $+x_E$ is north, $-y_E$ east, and $+z_E$ up; the initial structural base is $[0, 0, R_E]^\mathsf{T}$ with $R_E = 6371000\,\mathrm{m}$.

The scalar-first Hamilton quaternion $\boldsymbol{q}_{BE} = [q_w, q_x, q_y, q_z]^\mathsf{T}$ maps Body components into E using $\mathbf{C}_{BE}$. Equation (1) defines the vector transformation and launch orientation:

$$\boldsymbol{u}^E = \mathbf{C}_{BE}\boldsymbol{u}^B, \qquad \boldsymbol{q}_{BE}(0) = \begin{bmatrix} \sqrt{1/2} & 0 & -\sqrt{1/2} & 0 \end{bmatrix}^\mathsf{T}, \qquad \mathbf{C}_{BE}(0) = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}. \tag{1}$$

Thus Body $+x$ is initially upward, Body $+y$ maps to $+y_E$, and Body $+z$ maps to $-x_E$. Quaternions are normalized, and $\boldsymbol{q}$ and $-\boldsymbol{q}$ denote the same orientation. The adopted latitude $\lambda = 28.6°$ rotates the Earth-rate vector into the launch-aligned frame as $\boldsymbol{\Omega}^E = \Omega_E[\cos\lambda, 0, \sin\lambda]^\mathsf{T}$, with $\Omega_E = 7.292115 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$. It does not relocate the pad into Greenwich-axis coordinates.

Reference point and component basis are separate choices. Table 2 distinguishes them. A structural station $\boldsymbol{s}^B$ is measured from the stage base, whereas the corresponding CG force arm is $\boldsymbol{d}^B = \boldsymbol{s}^B - \boldsymbol{c}^B$, where $\boldsymbol{c}^B$ is the current CG station.

Quantity	Meaning
Dynamic position $\boldsymbol{r}_C^B$	Earth centre to instantaneous CG, expressed in current Body axes.
Dynamic velocity $\boldsymbol{v}_C^B$	CG velocity relative to rotating Earth, in current Body components.
Vehicle-object position/velocity	Structural-base state, with the same external origin and Body component basis.
Component/sensor station $\boldsymbol{s}^B$	Fixed structural location measured from the relevant stage base.
Moment and inertia	About the instantaneous vehicle CG, in Body axes.
Filter position/velocity	At the fixed IMU station, expressed in frame E.

Table 2 Shared origins and representations. Storing an Earth-centred vector in Body components does not make its origin the CG.

2.3 State and load interfaces

The physical state stores 15 numbers: CG position and velocity in Body components, a four-component quaternion, Body angular rate, and two stage-indexed propellant inventories. Each child retains this layout after separation, with the detached inventory set to zero. The estimator instead tracks the fixed IMU station in E. Its nominal/error-state definitions belong to Section 7; sensor stations and rate conventions belong to Section 6. Comparisons always use matching points, frames and epochs.

The load record separates nongravitational force, CG moment, central gravity, rotating-frame acceleration, stage consumption and nozzle-exit flow data. A station force contributes $(\boldsymbol{s} - \boldsymbol{c}) \times \boldsymbol{F}$ once; free moments are supplied separately. Gravity and frame acceleration enter the dynamics once as accelerations. This contract lets environment, mass, dynamics and navigation models evolve without redefining shared variables.

Trial evaluations recompute loads and properties without advancing accepted inventories, event clocks or sensor random generators. Only accepted states produce measurements and output. Section 8 describes the execution order, step boundaries and child propagation; Section 9 evaluates the resulting flight and navigation histories. Return commands currently use physical-state feedback, so landing performance and navigation accuracy are separate experiments.

3 Vehicle Configuration, Dynamics, and Mission Definition

3.1 Configuration and evidence-qualified parameters

The attached vehicle comprises stage 1, stage 2, and a loaded Dragon/trunk assembly. After separation, the lower vehicle contains stage 1 and the upper vehicle contains stage 2 and Dragon. The parameter register distinguishes K (authoritative published information), E (literature/data-derived estimate), and A (explicit modeling assumption). E/A denotes a sourced estimate whose use also requires an assumption. Source authority and applicability to the selected 2017 flight are separate considerations.

Table 3 consolidates the configuration used by `vehicle_parameters.py` and `VehicleData.py`. The 62.4 m equivalent height is the sum of adopted stage/Dragon lengths; it is not presented as a measured CRS-10 overall dimension. Stage boundaries and the contiguous attachment datums remain conditional. The 8430 kg loaded Dragon mass is retained as a provisional external estimate; the NASA cargo inventory is included within it. No separate satellite fairing is added.

Parameter	Adopted value	Class	Basis
Outer diameter	3.66 m	K	Later official external geometry [5].
Stage-1 / stage-2 length	42.6/12.6 m	E/A	Legacy FT geometry and assumed station allocation [17].
Stage-1 dry / propellant mass	22200/410900 kg	E	Adopted research mass budget [7].
Stage-2 dry / propellant mass	4000/107500 kg	E	Same literature set [7].
Loaded Dragon assembly	8430 kg	E	Provisional external estimate [3].
Dragon equivalent length / diameter	7.2/3.66 m	E/A	Earlier Dragon geometry transferred to a uniform assembly [4].
Dragon / stage-2 CG station	58.8/48.9 m	A	Midpoints above assumed attached-stage datums.
Stage-1 nominal dry CG	17.521 m	E/A	Component budget and residual allocation.
Initial attached mass	553030 kg	E/A	Sum of the adopted inventories and structures.
Equivalent model height	62.4 m	E/A	Sum of the adopted axial envelopes.

Table 3 Consolidated vehicle inputs. Stations are measured from the attached stack base along Body $+x$. A known external diameter does not establish a known internal tank or mass distribution.

3.2 Propulsion and propellant depletion

Stage 1 has one central engine and eight surrounding engines on an assumed 1.3 m radius ring. Stage 2 has one MVac engine. The force-application pivots are 1.0 m and 0.5 m above the respective bases (A); pivot stations, engine mass-centre stations, and nozzle-exit stations are distinct. Engine dry masses are already included in the stage dry budgets.

For a running steady engine with throttle u, the implementation evaluates Equation (2); an off engine has zero thrust and flow:

$$T = uT_{\mathrm{vac}} - A_e p_{\mathrm{amb}}, \qquad Q = \frac{uT_{\mathrm{vac}}}{I_{\mathrm{sp,vac}} g_0}, \qquad g_0 = 9.80665\,\mathrm{m\,s^{-2}}. \tag{2}$$

Here T is thrust, Q positive propellant outflow, A_e nozzle exit area, p_{amb} ambient pressure, and $I_{\mathrm{sp,vac}}$ the adopted vacuum specific impulse. The code retains a fixed-area ambient-pressure correction; the steady throttle does not scale that pressure term. Smooth ignition/shutdown envelopes subsequently scale the evaluated thrust and flow together. The effective stage-1 vacuum point is $756000 + 0.68(101325) = 824901$ N at full throttle, using the historical sea-level reference and a literature exit area. This mixed-source operating set is E/A, not a certified CRS-10 engine rating [6, 7]. Table 4 lists the adopted engine inputs.

Parameter	Stage-1 engine	Stage-2 engine
Engine count	9	1
Adopted vacuum thrust	824.901 kN (E/A)	935 kN (E)
Vacuum specific impulse	311 s (E)	348 s (E)
Exit area	0.68 m^2 (E)	4.90 m^2 (E)
Steady minimum throttle	0.60 (E)	0.40 (E)

Table 4 Engine operating inputs [7, 6]. The numerical scenario chooses engine count and throttle separately from these engine relations. A stored 5° gimbal parameter (A) does not establish implemented physical gimbal dynamics.

Stage-1 loading is $m_{f,0} = 123500\,\text{kg}$ of RP-1 and $m_{o,0} = 287400\,\text{kg}$ of LOX [9]. Their loaded ratio is 2.32713, while consumption uses $\eta = Q_o/Q_f = 2.35$ [8]. Equation (3) reconstructs both tanks from the stage's total consumed mass Δm:

$$Q_f = \frac{Q}{1+\eta}, \quad Q_o = \frac{\eta Q}{1+\eta}, \qquad m_f = m_{f,0} - \frac{\Delta m}{1+\eta}, \quad m_o = m_{o,0} - \frac{\eta \Delta m}{1+\eta}. \tag{3}$$

The adopted combination depletes LOX first and leaves 1202.128 kg of RP-1 at its mathematical limit. That residual is unusable under the assumed fixed ratio and is separate from a mission reserve. Reconstruction assumes fixed initial loading and no venting, refilling, or variable mixture schedule. Stage 2 retains an equivalent uniformly distributed propellant cylinder rather than the first-stage dual-column model.

3.3 Changing mass, CG, and inertia

3.3.1 Liquid and dry-component models

Each first-stage liquid is a settled, flat-ended cylindrical column at constant density. Table 5 lists its geometry and reference conditions. Structural section lengths are transferred to equivalent tank envelopes; domes, internal hardware displacement, and datum alignment are assumptions.

Quantity	RP-1	LOX
Inner radius [m], E/A	1.8269	1.8271
Bottom station [m], E/A	4.0	18.7
Equivalent section length [m], E/A	14.7	22.1
Density [kg m^{-3}], E	820	1250
Reference temperature [K], E	266	66

Table 5 Equivalent tank inputs. Radii subtract the modeled skin thickness from 1.83 m; section lengths/stations use Ecker's structural model [15, Table 1]. Density/temperature pairs use Miquel's Table 5, whose reference pressure is unspecified [16].

For liquid mass m_i, density ρ_i, radius r_i, bottom station b_i, and outflow Q_i, Equation (4) defines height, centroid, and centroid rate:

$$h_i = \frac{m_i}{\rho_i \pi r_i^2}, \qquad c_i = b_i + \frac{h_i}{2}, \qquad \dot{c}_i = -\frac{Q_i}{2\rho_i \pi r_i^2}. \tag{4}$$

The height must remain inside the adopted envelope. Slosh, ullage motion, reverse-acceleration settling, and feed transients are not resolved. The settled-column approximation is retained through coast and return; restart settling is a separate scenario assumption.

The dry allocation is shell 8000 kg at 21.0 m, fins 1520 kg at 40.3 m, engines 4230 kg at 0.452 m, and folded legs 2000 kg at 6.66 m, using published group estimates with equivalent shapes [8, Table 2]. The remaining 6450 kg closes the 22200 kg budget and is assumed uniformly distributed over 4.0–40.8 m. This yields nominal dry CG 17.521 m. Moving the residual group's centroid anywhere within that interval gives conditional dry-CG bounds 12.18–22.87 m; they are allocation bounds, not statistical confidence limits.

The shell/residual groups use thin-shell equivalents and localized groups use equivalent disks. The liquid columns and loaded Dragon use uniform solid cylinders. Dry-group placements stay fixed; deployed aerodynamic/contact geometry does not separately relocate the dry leg mass. The initial combined vehicle has CG

station 29.3704 m and inertia diagonal approximately $[9.47993 \times 10^5, 1.13857 \times 10^8, 1.13857 \times 10^8]\,\mathrm{kg\,m^2}$. These are model outputs (E/A), not measured CRS-10 mass properties.

3.3.2 Component aggregation and analytical derivatives

A `MassComponent` describes one part or liquid about its own centre. `MassProperties` describes the attached vehicle about its current CG. With component mass m_i, axial centroid c_i, local inertia $\mathbf{J}_{i,c}$, and offset $\boldsymbol{d}_i = (c_i - c)\boldsymbol{e}_x$, Equation (5) gives the combined properties:

$$\begin{aligned} m &= \sum_i m_i, \qquad c = \frac{\sum_i m_i c_i}{m}, \\ \mathbf{J}_C &= \sum_i \left[\mathbf{J}_{i,c} + m_i \mathbf{S}(\boldsymbol{d}_i)\right], \qquad \mathbf{S}(\boldsymbol{d}) = (\boldsymbol{d}^\top \boldsymbol{d})\mathbf{I} - \boldsymbol{d}\boldsymbol{d}^\top. \end{aligned} \tag{5}$$

Here $\mathbf{I}$ is the identity matrix. The implementation differentiates both component properties and parallel-axis offsets. Equation (6) states the first derivatives; second CG/mass derivatives are also evaluated analytically for the motion and sensor interfaces:

$$\begin{aligned} \dot{c} &= \frac{\sum_i (\dot{m}_i c_i + m_i \dot{c}_i) - \dot{m} c}{m}, \\ \dot{\mathbf{J}}_C &= \sum_i \left[\dot{\mathbf{J}}_{i,c} + \dot{m}_i \mathbf{S}(\boldsymbol{d}_i) + m_i \dot{\mathbf{S}}(\boldsymbol{d}_i)\right]. \end{aligned} \tag{6}$$

Axial placement makes the parallel-axis addition $\mathrm{diag}(0, d_i^2, d_i^2)$. Thus Body I_{xx} is axial and I_{yy}, I_{zz} are transverse. Figure 3 shows this same operation in code, and Figure 4 shows its first-stage-only inventory dependence. The late change in CG trend follows the changing weights of the two liquid centroids and the fixed dry allocation.

mass_properties.py | lines 157-169

```
157     inertia = np.zeros((3, 3))
158     inertia_rate = np.zeros((3, 3))
159     for c in components:
160         distance = c.center_axial_m - center
161         distance_rate = c.center_rate_mps - center_rate
162         parallel = np.diag([0.0, distance**2, distance**2])
163         parallel_rate = np.diag([0.0] + [2 * distance * distance_rate] * 2)
164         inertia += c.inertia_at_center_kgm2 + c.mass_kg * parallel
165         inertia_rate += (
166             c.inertia_rate_at_center_kgm2ps
167             + c.mass_rate_kgps * parallel
168             + c.mass_kg * parallel_rate
169         )
```

Figure 3 Exact parallel-axis and derivative accumulation in `mass_properties.py`. Each local inertia is shifted to the combined CG once. Source line numbers and file hash are retained with the figure provenance.

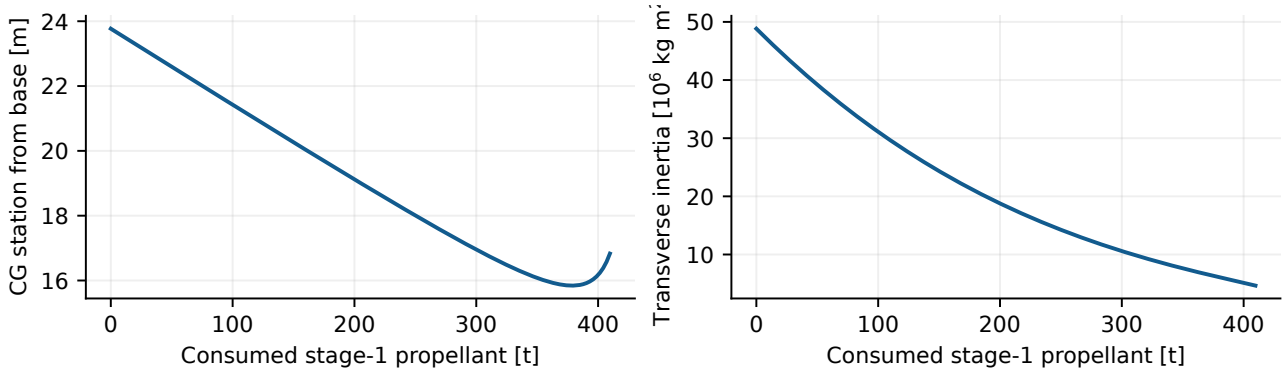


Figure 4 Stage-1-only mass-property sweep from full loading to the fixed-ratio limiting inventory. The curves evaluate the adopted component model against consumed propellant; they are neither measured flight properties nor a separate trajectory simulation.

3.4 Equations of motion and load closure

The changing-mass model uses fixed nozzle-exit stations and balanced transverse outflow. For outlet j, $Q_j \geq 0$ is outward flow, $\mu_j = -Q_j$ its signed mass-removal rate, $\dot{\mu}_j = -\dot{Q}_j$, and $\boldsymbol{d}_j = \boldsymbol{s}_j - \boldsymbol{c}$ the exit arm. $\mathbf{G}_j$ is the outlet's second-moment-per-unit-mass tensor in m^2. The open-system balance retains explicit flow terms, consistent with control-volume rocket dynamics [18].

Equations (7) and (8) are the implemented CG balances. $\boldsymbol{a}_C^B$ is the external-frame coordinate acceleration expressed in Body components; $\boldsymbol{g}^B$ and $\boldsymbol{a}_{\mathrm{rot}}^B$ are supplied separately by the environment:

$$m\boldsymbol{a}_C^B = \boldsymbol{F}^B + m\boldsymbol{g}^B + m\boldsymbol{a}_{\mathrm{rot}}^B + \sum_j \left[2\mu_j(\boldsymbol{\omega}^B \times \boldsymbol{d}_j^B - \dot{\boldsymbol{c}}^B) + \dot{\mu}_j \boldsymbol{d}_j^B\right], \tag{7}$$

$$\mathbf{J}_C\dot{\boldsymbol{\omega}}^B = \boldsymbol{M}_C^B - \boldsymbol{\omega}^B \times (\mathbf{J}_C\boldsymbol{\omega}^B) - \dot{\mathbf{J}}_C\boldsymbol{\omega}^B + \sum_j \mu_j[\mathbf{S}(\boldsymbol{d}_j^B) + \mathbf{G}_j]\boldsymbol{\omega}^B. \tag{8}$$

Pressure-corrected thrust already contains the exhaust reaction; another Qv_e force is not added. The remaining flow terms describe reference migration and angular-momentum outflow. The current closure requires axial CG migration, fixed exits, and balanced transverse exhaust centroids; internal swirl and moving nozzle boundaries are omitted. The code verifies consistency between consumption and mass derivatives before evaluating these balances.

Because the state is stored in current Body axes, Equation (9) includes basis-transport terms. The quaternion uses the same approximate relative-rate convention as Section 2:

$$\begin{aligned} \dot{\boldsymbol{r}}_C^B &= \boldsymbol{v}_C^B - \boldsymbol{\omega}^B \times \boldsymbol{r}_C^B, & \dot{\boldsymbol{v}}_C^B &= \boldsymbol{a}_C^B - \boldsymbol{\omega}^B \times \boldsymbol{v}_C^B, \\ \dot{\boldsymbol{q}}_{BE} &= \tfrac{1}{2}\boldsymbol{q}_{BE} \otimes [0, (\boldsymbol{\omega}^B)^{\mathsf{T}}]^{\mathsf{T}}, & \dot{m}_{p,k} &= -Q_k. \end{aligned} \tag{9}$$

The integrator temporarily uses external translational components within RK4, then returns the accepted state in Body components. This avoids artificial radial interpolation error from rotating an Earth-scale stored vector.

Structural-base and CG states are related by Equation (10), with base O and current CG station $\boldsymbol{c}^B$:

$$\boldsymbol{r}_C^B = \boldsymbol{r}_O^B + \boldsymbol{c}^B, \qquad \boldsymbol{v}_C^B = \boldsymbol{v}_O^B + \boldsymbol{\omega}^B \times \boldsymbol{c}^B + \dot{\boldsymbol{c}}^B. \tag{10}$$

The same transformation applies to a fixed sensor station after replacing the base-to-CG displacement appropriately. `to_state` and `apply_state` use matching flow/rate inputs. Their initial stationary-base condition can therefore have a nonzero geometric-CG velocity from depletion, without implying motion of the sensor mount.

Control moments are supplied inputs to these equations. Their idealized authority and physical actuator limitations are defined in Section 8.

3.5 Initial state and flight-phase definition

The structural base starts at the launch location of Section 2, with zero velocity relative to rotating Earth and zero Body-relative angular rate. The launch quaternion sets Body $+x$ upward; inventories are the full values in Table 3. Commands, configuration transitions, and continuous integration then generate all subsequent states.

Table 6 defines the mission sequence without substituting simulated event times for input conditions. The official schedule labels its entries approximate. Adopting a published time as a simulation command is an A-class choice even when the published schedule itself is authoritative [1].

First contact uses four assumed leg tips 1.0 m below the base at 3.5 m radius. A successful contact additionally requires deployed legs, vertical speed at most $2\,\mathrm{m\,s^{-1}}$, horizontal speed at most $1\,\mathrm{m\,s^{-1}}$, tilt at most 5°, angular-rate magnitude at most $1\,\mathrm{deg\,s^{-1}}$, and target error at most 10 m. These are A-class project limits, not certified Falcon 9 requirements. The target is an assumed point 14 km south and 6 km east of launch on the ground sphere, rather than surveyed LZ-1 coordinates.

For normalized ramp time $\xi = (t - t_s)/\Delta t$, Equation (11) supplies a smooth shutdown amplitude and its time derivative on $0 \leq \xi \leq 1$:

$$\gamma = 1 - 10\xi^3 + 15\xi^4 - 6\xi^5, \qquad \dot{\gamma} = -\frac{30\xi^2(1-\xi)^2}{\Delta t}, \qquad \Delta t = 1\,\mathrm{s}. \tag{11}$$

The envelope is constant outside the ramp, with zero endpoint slope and curvature. Scaling thrust and flow together while supplying the analytical flow derivative avoids an unmodeled discontinuous-flow impulse. The duration is a numerical assumption, not a measured engine transient. Return planning budgets, steering laws, and actuator allocation are kept in mission execution rather than embedded in the vehicle equations.

Transition	Definition in the selected scenario
Liftoff / ascent	Initial attached vehicle; prescribed ascent commands.
Shutdown begins	140 s; project one-second smooth ramp.
MECO / stack coast	141 s; thrust and flow reach zero.
Stage separation	144 s; attached parent, engines off, zero flow and flow derivative.
Upper-stage ignition	152 s; independent upper-child commands.
Boostback ignition	161 s; detached booster and fuel/attitude/readiness guards.
Boostback end / return coast	Miss, fuel, or duration condition followed by smooth shutdown.
Entry ignition	Descending through 45 km with restart guards.
Entry end / aerodynamic descent	$500\,\mathrm{m\,s^{-1}}$ speed target or planning cap, then shutdown.
Landing ignition	Project stopping-clearance prediction with restart guards.
First contact	Lowest assumed deployed leg tip reaches the ground sphere; evaluate landing criteria.
Upper-stage cutoff	545 s; prescribed upper-child shutdown.

Table 6 Physical phase definitions and command references. Entry/landing conditions and return planning criteria are project assumptions. Their actual execution times and numerical results belong to the simulation-results analysis.

3.6 Separation and inherited child states

The separation map assumes zero impulse and no discarded mass. Both children inherit the parent's attitude and angular rate. For a cut offset $\boldsymbol{\ell}^B$ from the parent structural base, the lower base retains the parent's base state and the upper base follows Equation (12):

$$\boldsymbol{r}^B_{O,u} = \boldsymbol{r}^B_{O,p} + \boldsymbol{\ell}^B, \qquad \boldsymbol{v}^B_{O,u} = \boldsymbol{v}^B_{O,p} + \boldsymbol{\omega}^B \times \boldsymbol{\ell}^B. \tag{12}$$

Upper-child component stations are rebased consistently, while Dragon stays on that child. Different child CG coordinates are expected because the tracked points have changed. Position/velocity continuity is checked at fixed structural points and at the recombined mass centre.

The conservation audit uses angular momentum about the parent CG. With child CG offset $\boldsymbol{\rho}_k$ and velocity difference $\Delta\boldsymbol{v}_k$ from the parent CG, Equation (13) includes spin and displaced-CG motion:

$$\boldsymbol{H}_{\text{after}} = \sum_k \left[\mathbf{J}_k\boldsymbol{\omega}_k + \boldsymbol{\rho}_k \times (m_k\Delta\boldsymbol{v}_k)\right]. \tag{13}$$

The map compares this value with the parent spin angular momentum, and also checks mass, linear momentum, kinetic energy, recombined CG, and structural-point continuity. Relative conservation tolerance is 10^{-10}; position and velocity continuity tolerances are $10^{-8}\,\mathrm{m}$ and $10^{-8}\,\mathrm{m\,s^{-1}}$; attitude/rate checks use 10^{-12} in their respective norms. These are numerical consistency criteria for the assumed map, not physical separation accuracy.

Afterward, stage 1 excludes stage 2 and Dragon from its mass properties, and the upper child excludes stage 1. Both use inherited states and separate subsequent commands on the same clock. The upper stage retains the uniform propellant/Dragon approximations and its separate drag provider, rather than automatically selecting the declared `second_stack` aerodynamic configuration. Dragon remains attached over the 600 s demonstration interval. The sequence tests continuous configuration/state handling; public validation of the complete return and upper-stage command laws is outside the available ascent reference.

4 Environment Models

The environment module (`Environement.py`) is a pure service layer: it supplies the world the vehicle flies through and never calls back into vehicle, dynamics or control code. Its services are the standard atmosphere, point-mass gravity with its gradient, the acceleration of the rotating Earth-fixed frame, the wind, and the ground reference; all evaluations are stateless functions of the current state. Following the project convention, any state outside a modeled domain is rejected with an error, never extrapolated.

4.1 Atmosphere

The atmosphere follows the US Standard Atmosphere 1976 [10]: seven gradient layers between sea level and the mesopause at 84.852 km, reproduced layer by layer. Within a layer with base temperature T_b, base pressure p_b and lapse rate L, and in the isothermal case, the pressure is given by Equation (14):

$$p(h) = p_b \left(\frac{T_b + L\,(h - h_b)}{T_b} \right)^{-\frac{g_0}{R\,L}} \quad (L \neq 0), \qquad p(h) = p_b \, \exp\left(-\frac{g_0\,(h - h_b)}{R\,T_b} \right) \quad (L = 0), \tag{14}$$

with $g_0 = 9.80665\,\mathrm{m/s^2}$ and the gas constant of dry air $R = 287.05287\,\mathrm{J/(kg\,K)}$. Density and speed of sound follow the ideal-gas and isentropic relations, $\rho = p/(R\,T)$ and $a = \sqrt{\gamma R T}$ with $\gamma = 1.4$. Sea-level values are $T = 288.15\,\mathrm{K}$, $p = 101\,325\,\mathrm{Pa}$, $\rho = 1.225\,\mathrm{kg/m^3}$.

Above the mesopause the model deliberately stops claiming physics: an isothermal tail at $T_m = 186.946\,\mathrm{K}$ decays exponentially with a scale height of $R\,T_m/g_0 \approx 5.47\,\mathrm{km}$ toward numerical vacuum (density below $10^{-150}\,\mathrm{kg/m^3}$ at the 2 000 km envelope ceiling; $p(200\,\mathrm{km}) \approx 2.7 \times 10^{-10}\,\mathrm{Pa}$). The envelope 0–2 000 km is the mission scope; altitudes outside raise a domain error instead of a number. Figure 5 shows the model against the published USSA 1976 checkpoint values that the test suite pins (5–80 km, temperatures, pressures and densities at 5×10^{-5} relative tolerance); the environment suite of 41 tests additionally checks layer-boundary continuity, monotone decay, exact tail scale height, hydrostatic balance and the domain rejection behavior.

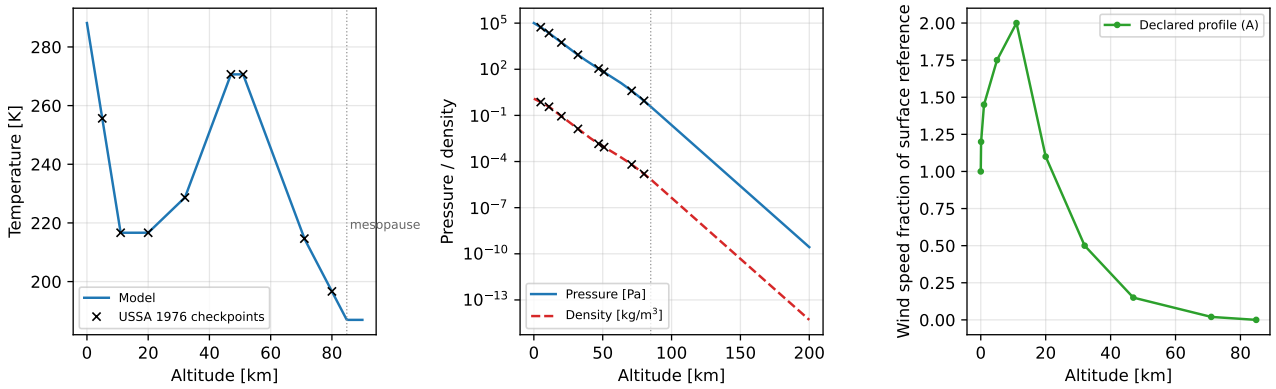


Figure 5 Environment model. Left: layer temperatures to the mesopause with published USSA 1976 checkpoint values. Center: pressure and density (logarithmic) to 200 km, showing the isothermal tail above the mesopause; black markers are the published checkpoints. Right: the declared wind-profile scale factor of the surface reference wind (class A), which is zero above the modeled atmosphere; the nominal mission runs in calm air.

4.2 Gravity and rotating-frame accelerations

Gravity is the reduced Newtonian point-mass model. With the Earth radius $R_E = 6\,371\,\mathrm{km}$ and the configured surface gravity g_0, the gravitational parameter is $\mu = g_0 R_E^2$, so the gravitational acceleration at the launch site is the configured value by construction:

$$\mathbf{a}_g = -\frac{\mu}{\|\mathbf{r}\|^3}\,\mathbf{r}. \tag{15}$$

Gravity never appears as a force; Equation (15) is added once as an acceleration next to thrust over mass. The module also provides the 3×3 gravity-gradient map $\nabla_{\mathbf{r}}\mathbf{a}_g = (\mu/\|\mathbf{r}\|^3)\,(3\,\hat{\mathbf{r}}\hat{\mathbf{r}}^{\mathsf{T}} - \mathbf{I})$, used only in the navigation filter's prediction Jacobian (Section 7), not for torques. Oblateness (J_2) is out of scope.

The external frame is Earth-centred and launch-aligned (the shared conventions are fixed in the architecture chapter of this report); it rotates with Earth relative to inertial space. Translation uses the separate frame acceleration in Equation (16)

$$\mathbf{a}_{\mathrm{fr}} = -2\,\boldsymbol{\omega}_E \times \mathbf{v} \;-\; \boldsymbol{\omega}_E \times (\boldsymbol{\omega}_E \times \mathbf{r}), \tag{16}$$

the Coriolis and centrifugal terms for velocity relative to the rotating Earth, with the rotation vector $\boldsymbol{\omega}_E = \Omega\,(\cos\varphi,\ 0,\ \sin\varphi)$ at launch latitude φ. The nominal mission uses the sidereal rate $\Omega = 7.2921150 \times 10^{-5}$ rad/s at $\varphi = 28.6°$; setting $\Omega = 0$ disables the terms. The frame accelerations, and their analytical gradients, are supplied once per right-hand-side evaluation next to gravity. The angular equations and the gyro model retain the project's approximate Earth-rate treatment: no complete inertial-rate coupling is claimed.

4.3 Wind and ground reference

The wind model is a surface reference vector (ECEF components in the mission configuration) that can be rotated about the launch-site vertical, scaled by a declared altitude profile: the scale factor of the surface reference grows through the boundary layer, peaks at 2.0 around the 11 km jet-stream core, decays through the stratosphere and reaches exactly zero at the mesopause — there is no wind above the modeled atmosphere (Figure 5, right). The profile is a declared stylization (class A). There are no gusts, no time dependence and no direction shear, and the wind enters the physics only through the relative-velocity term of the aerodynamics model. The nominal full-mission run uses calm air: the mission configuration sets no surface reference wind, so the default zero vector applies throughout.

The ground is a bare sphere of radius R_E; altitude is measured radially above it, with an optional reference-level shift, and supports a station offset from the center of gravity (used, for example, for engine-exit altitude). Terrain and contact forces are out of scope; contact is evaluated at the mission level through the leg-tip geometry discussed with the landing criteria in the mission-execution chapter of this report.

Assumptions and limitations The environment is dry air only, with a strictly layered USSA profile; gravity is spherical with no oblateness; the wind is a declared stylization without gusts, time dependence or direction shear; the ground is a bare sphere without terrain; and the frame-acceleration treatment is exact for translation while the attitude chain carries only the approximate Earth-rate treatment noted above. These scope choices (class A) apply uniformly to ascent and return; no environment parameter was fitted to flight data. The module validates its configuration at construction — non-finite gravity, radius, ground offset or wind, non-positive radius, a negative rotation rate or a latitude beyond ±90° all raise an error before flight, not mid-integration.

5 Aerodynamics

`Aerodynamics.py` returns Body-frame forces and CG moments using the evidence-qualified register `data/aero_configurations.json`. Four configurations serve stack/booster phases; a fifth, `second_stack`, is declared but not selected by the independent upper-stage drag provider (Section 8).

5.1 Flow state and force build-up

Equation (17) defines the relative flow and total angle:

$$\mathbf{v}_{\text{rel}} = \mathbf{v}_{\text{CG}} - \dot{\mathbf{c}} - \mathbf{w}, \qquad \alpha = \operatorname{atan2}\big(\|\mathbf{v}_{\text{rel},yz}\|,\ v_{\text{rel},x} \big), \tag{17}$$

where $\dot{\mathbf{c}}$ is the axial center-of-gravity migration rate (Section 3, stage depletion) and $\mathbf{w}$ the wind in Body components (Section 4). The total flow angle α spans 0 (nose-first) to π (engine-first); every direction in the lateral plane is handled, not only a pitch plane. The dynamic pressure is $\bar{q} = \frac{1}{2}\,\rho\, v_{\text{rel}}^2$ with the atmosphere state of Section 4.

The axial and potential-normal loads follow Equation (18); side segments add the crossflow terms of Equation (19):

$$F_x = -\,\bar{q}\, S_{\text{ref}}\, CA_0(M)\ \cos\alpha\,|\cos\alpha|, \qquad N = \bar{q}\, S_{\text{ref}}\, C_{N\alpha}\ \sin\alpha\,|\cos\alpha|, \tag{18}$$

$$D_i = \bar{q}\ C_{Dc,i}\, S_i\ \sin^2\alpha \qquad \text{per side segment } i, \tag{19}$$

Reference areas/lengths appear in Table 7. Normal force opposes lateral flow at a windward-end station (57.6 m for the stack, 41.0 m for the booster), mirrored about CG for retro flow. Crossflow forces act at side-area centroids. The landing configuration adds fin and deployed-leg segments to the body model.

The $\cos\alpha|\cos\alpha|$ factor reverses the axial load and fixed-deflection fin effect in engine-first flight and makes them vanish at 90°. Station moments are assembled once as $(\boldsymbol{s} - \boldsymbol{c}) \times \boldsymbol{F}$. Geometry determines restoring or destabilizing effects; the model does not impose passive stability.

Equation (20) closes the moment set with per-axis damping:

$$\mathbf{M}_{\text{damp}} = \operatorname{diag}(c_l, c_m, c_m)\ \frac{\bar{q}\, S_{\text{ref}}\, L_{\text{ref}}^2}{2\, v_{\text{rel}}}\ \boldsymbol{\omega}, \tag{20}$$

with pitch and yaw coefficient c_m tied by symmetry and roll coefficient c_l separate (declared values: -10 full stack, -20 booster, roll -3 with fins deployed; a zero coefficient means that no roll-damping contribution is applied).

5.2 Configurations and axial curves

Configuration	Phase	S_{ref} [m^2]	L_{ref} [m]	Axial curve / fins
`full_stack`	launch to separation	21.08	62.4	sourced M 0–7; declared M 7–10
`booster_alone`	coast, boostback	10.52	42.6	declared (±50 %)
`booster_return`	entry burn, descent	10.52	42.6	declared; four fins
`booster_landing`	landing burn	10.52	42.6	declared; four fins and legs
`second_stack`	stage-2 body (declared)	10.52	19.8	declared, sleeker estimate

Table 7 The five aerodynamic configurations. Station datum is the configuration's own structural base. Fins deploy after boostback; the legs deploy at landing ignition, which is represented by the segment difference between `booster_return` and `booster_landing`.

Figure 6 shows the adopted $CA_0(M)$ curves. The full-stack curve to Mach 7 is read from Pezent's RasAeroII analysis of a Falcon-9-like stack with fairing [7]; its transonic peak is about 1.16 near Mach 1.06. The extension to Mach 10 and all booster/upper curves are engineering estimates, with declared uncertainties around ±50% for the unsourced curves. The fairing-stack curve is transferred to Dragon geometry as an E/A approximation: reference area and coefficient normalization must be retained together. It is not a CRS-10-specific aerodynamic identification.

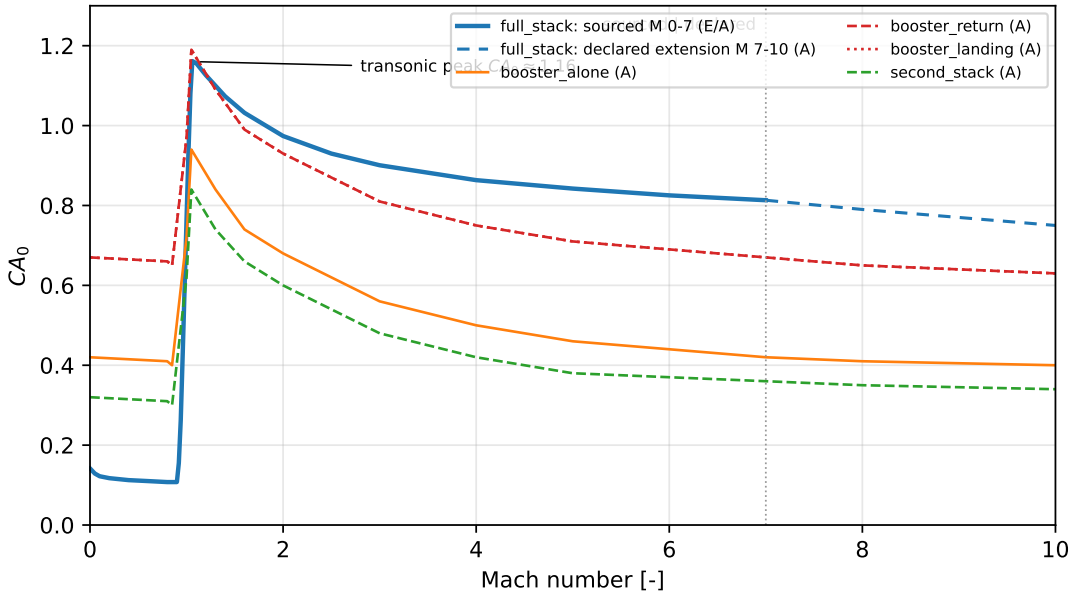


Figure 6 Axial drag coefficient $CA_0(M)$ of the five configurations. Only the full-stack curve to Mach 7 is sourced; everything else is a declared estimate (evidence classes in the legend). The transonic peak of the full stack reaches $CA_0 \approx 1.16$.

5.3 Grid fin control

The return/landing configurations have four fins at 45°, 135°, 225° and 315° about Body $+x$, station 39.6 m and pivot radius 1.83 m. Each uses area 1.65 m^2, effective lift slope 1.2 rad^{-1} and travel ±0.35 rad, all research-model assumptions. Equation (21) gives the force for deflection δ_i and azimuth φ_i:

$$\boldsymbol{F}_i = \bar{q} S_{\text{fin}} c_{\ell,\delta} \delta_i \cos\alpha |\cos\alpha| \begin{bmatrix} 0 & -\sin\varphi_i & \cos\varphi_i \end{bmatrix}^{\mathsf{T}}. \tag{21}$$

The three-dimensional mount arm generates pitch/yaw through axial offset and roll through radial offset. Equal deflections produce roll; differential pairs generate transverse moments. Command selection belongs to Section 8. The aerodynamic interface applies instantaneous deflections without servo dynamics. Mach/angle-dependent effectiveness, plume interference and deflection-induced axial drag are unresolved.

5.4 Domain and verification

The model rejects unsupported Mach values (the curves end at 10), unknown configurations, invalid stations and out-of-travel commands. Declared coefficient uncertainty remains an uncertainty within that domain, not an execution error.

The 42 aerodynamic tests check axial baselines, force direction, retro reversal, moment arms, dissipative damping, continuity, fin combinations and all configuration records, plus a short propagation through the dynamics provider. These checks verify the implementation; they do not validate the unsourced coefficient curves against flight data.

6 Sensor Models and Measurement Generation

Four sensor front-ends generate measurements from the truth trajectory: an inertial measurement unit (IMU), a global navigation satellite system (GNSS) receiver, a light detection and ranging (LiDAR) altimeter, and a star tracker (`sensors.py`). Sampling occurs only at accepted integration times. The nominal sensor periods are integer multiples of the 0.01 s maximum step, so all rates in Table 8 are hit exactly; no random numbers are drawn during integration trial evaluations, and a fixed seed reproduces a run sample for sample. The navigation-consumer side of these streams is the filter of Section 7; this section covers measurement generation only. All specs are placeholder research assumptions (class A), not certified Falcon 9 hardware.

Sensor	Rate	Mounting	Error model
IMU	100 Hz	station $[38, 0, 0]$	accel $\sigma = 0.01\,\mathrm{m/s^2}$, bias $[0.05, -0.03, 0.02]\,\mathrm{m/s^2}$, scale factor 10^{-3}, bias random walk $10^{-4}\,\mathrm{m\,s^{-5/2}}$; gyro $\sigma = 10^{-4}\,\mathrm{rad/s}$, bias $[0, 0, 10^{-4}]\,\mathrm{rad/s}$, bias random walk $10^{-7}\,\mathrm{rad\,s^{-3/2}}$; misalignment and temperature models supported
GNSS	10 Hz	antenna $[41, 0, 0]$	pos $\sigma = [1.5, 1.5, 2.5]$ m plus Gauss-Markov $\sigma = [0.6, 0.6, 1.0]$ m at $\tau = 300$ s; vel $\sigma = 0.1$ m/s; latency 0.1 s; outages and dropouts supported
LiDAR	10 Hz	station $[0.5, 0, 0]$, boresight $-x$	slant range 1–1 800 m, $\sigma = 0.05\,\mathrm{m} + 10^{-4}\,R$
Star tracker	4 Hz	boresight 60° from $+x$ toward $+y$	cross-boresight $\sigma = 5$ arcsec, boresight roll $\sigma = 40$ arcsec; latency 0.1 s

Table 8 Sensor front-ends and adopted settings; station coordinates are in metres in Body axes. Every entry is a placeholder research assumption (class A), not a certified Falcon 9 sensor.

6.1 Inertial measurement unit

The IMU measures specific force and angular rate at its fixed structural station. The ideal specific force includes the station kinematics of the dynamics core — the angular-rate and center-of-gravity-migration terms of moving the measurement point off the center of gravity — so the accelerometer resolves the modeled structural-point motion. Equation (22) gives the measured pair:

$$\begin{aligned} \mathbf{f}_{\mathrm{meas}} &= \mathbf{A}\,\mathbf{f}_{\mathrm{true}} + \mathbf{b}_a + \mathbf{b}_{a,T} + \mathbf{b}_{a,\mathrm{rw}} + \mathbf{n}_a, \\ \boldsymbol{\omega}_{\mathrm{meas}} &= \mathbf{G}\,\boldsymbol{\omega}_{\mathrm{true}} + \mathbf{b}_g + \mathbf{b}_{g,T} + \mathbf{b}_{g,\mathrm{rw}} + \mathbf{n}_g. \end{aligned} \tag{22}$$

with per-axis scale-factor/misalignment matrices $\mathbf{A}, \mathbf{G} = \mathbf{I} + \mathrm{diag}(\cdot) + \mathbf{M}$ (zero diagonal of $\mathbf{M}$), constant biases, temperature coefficients about a 20 °C reference, bias random walks driven per sample as $\mathcal{N}(0, \sigma_{\mathrm{rw}}^2\,\Delta t)$, and white noise at the sigmas of Table 8. The nominal mission uses the biases, scale factor and random walk listed in the table, with misalignment, temperature and gyro scale-factor terms at zero (modeled but switched off). The gyro measures the Body rate relative to the rotating Earth-centred Earth-fixed frame (ECEF) frame, consistent with the quaternion transport equation the navigation filter propagates. This is an A-class relative-rate convention: a physical inertial gyro would additionally include Earth rate transformed into Body axes. The star-tracker ECI conversion does not supply that missing gyro term.

6.2 GNSS, LiDAR and star tracker

The GNSS receiver provides antenna position and velocity in the external frame, transformed from the truth state with the full lever-arm kinematics (antenna station minus IMU station, angular-rate cross term and center-of-gravity migration). A 0.1 s latency model feeds a truth-side history buffer that is linearly interpolated at the delayed epoch, so the emitted sample is a measurement of the past state, not of the present one. The position error is a first-order Gauss-Markov process beside the white noise (updating per sample with the configured 300 s correlation time), modelling the slow, correlated component absent from naive white-noise GNSS error.

First-fix delays, scheduled outages and random dropouts are supported and reported through a validity flag; the nominal run has none of them, and every sample is valid.

The LiDAR measures slant range to the spherical ground along its $-x$ boresight. Validity is geometric: the analytic line–sphere intersection must exist, point downward and land inside the declared 1–1 800 m envelope; invalid geometry is reported as an invalid sample, never extrapolated. Range noise is $0.05\,\mathrm{m} + 10^{-4}R$.

The star tracker provides absolute attitude as a sensor-frame-to-Earth-centred inertial frame (ECI) quaternion. The ECI frame is defined to coincide with the launch-aligned ECEF frame at $t = 0$, and the Earth rotation carries the conversion, so the injection of rotation into the attitude chain is explicit rather than implicit. Noise is 5 arcsec cross-boresight and 40 arcsec about the boresight per exposure, with a 0.1 s latency applied on the attitude history. Availability is physical: the unit is blinded above $2^\circ/\mathrm{s}$ body rate, below 60 km altitude (dense atmosphere), inside the 20° Earth-limb exclusion, and optionally inside a 40° solar exclusion; first-fix, outage and dropout faults are supported (nominally off). The reported sample always carries its blinding reason, making the availability record auditable in the run output.

Assumptions and limitations All noise, bias and latency values are declared placeholders, not datasheet values; the GNSS model has no ionospheric, tropospheric or multipath terms; the LiDAR assumes the bare sphere of Section 4 as ground; the star tracker mounting is a placeholder that keeps the boresight sky-pointing for an upright vehicle, and vehicle-generated obscuration (plumes, stage interfaces) is not modeled.

7 Navigation Architecture and State Estimation

The MEKF in `kalman.py` estimates motion at the fixed IMU station from the streams defined in Section 6. It uses its own attitude during prediction and correction. Vehicle control continues to use physical-state feedback; the estimator is evaluated independently in Section 9.

7.1 Nominal state, error state, and initialization

The nominal state stores $\hat{\boldsymbol{x}} = [\hat{\boldsymbol{p}}, \hat{\boldsymbol{v}}, \hat{\boldsymbol{b}}_a, \hat{\boldsymbol{q}}, \hat{\boldsymbol{b}}_g, \hat{\boldsymbol{s}}_f, \hat{\boldsymbol{m}}]$: position and velocity in E, Body accelerometer bias, the Body-to-E quaternion, gyro bias, three scale factors, and six off-diagonal calibration terms. This is 25 stored scalars, including the four-component unit quaternion. The covariance instead represents the 24-component error state in Equation (23):

$$\delta\boldsymbol{x} = [\delta\boldsymbol{p}^\mathsf{T}, \delta\boldsymbol{v}^\mathsf{T}, \delta\boldsymbol{b}_a^\mathsf{T}, \delta\boldsymbol{\theta}^\mathsf{T}, \delta\boldsymbol{b}_g^\mathsf{T}, \delta\boldsymbol{s}_f^\mathsf{T}, \delta\boldsymbol{m}^\mathsf{T}]^\mathsf{T}, \qquad \boldsymbol{q}_{\text{true}} = \hat{\boldsymbol{q}} \otimes \operatorname{Exp}(\delta\boldsymbol{\theta}). \tag{23}$$

Here Exp maps a three-component rotation vector to a unit quaternion and $\mathcal{N}(\boldsymbol{q}) = \boldsymbol{q}/\|\boldsymbol{q}\|$ normalizes it. The attitude error is expressed in Body axes. Table 9 defines the initial uncertainty. Position starts with the fixed perturbation $[1, -1, 0.5]^\mathsf{T}$ m; velocity and mean attitude start at the launch state. This known initial alignment is an A-class prior, not a measured alignment solution.

Error block	Size	Units	Initial standard deviation
Position / velocity	3 + 3	m / m s^{-1}	2 / 0.05 per axis
Accelerometer bias	3	m s^{-2}	0.1 per axis
Attitude error	3	rad	0.01 per axis
Gyro bias	3	rad s^{-1}	10^{-3} per axis
Gyro scale factor	3	dimensionless	10^{-3} per axis
Gyro misalignment	6	rad (small-angle terms)	10^{-3} per term

Table 9 Default 24-dimensional error covariance. The nominal quaternion has four stored components but three independent attitude-error coordinates. All initialization settings are A-class assumptions.

7.2 IMU prediction and covariance propagation

The gyro calibration matrix is $\hat{\mathbf{G}} = \mathbf{I} + \operatorname{diag}(\hat{\boldsymbol{s}}_f) + \mathbf{M}(\hat{\boldsymbol{m}})$, where $\mathbf{M}$ has the six estimated off-diagonal terms. Equation (24) gives the nominal prediction. Superscripts $-$ and $+$ denote predicted and corrected estimates, respectively; $\overline{\boldsymbol{\omega}}$ is the mean of consecutive calibrated rates and $\mathbf{C}_{1/2}$ uses the mid-interval quaternion:

$$\begin{aligned}
\hat{\boldsymbol{\omega}} &= \hat{\mathbf{G}}^{-1}(\boldsymbol{\omega}_{\text{meas}} - \hat{\boldsymbol{b}}_g), & \hat{\boldsymbol{q}}_{k+1}^- &= \mathcal{N}\left[\hat{\boldsymbol{q}}_k^+ \otimes \operatorname{Exp}(\overline{\boldsymbol{\omega}}\Delta t)\right], \\
\hat{\boldsymbol{a}}_k &= \mathbf{C}_{1/2}(\boldsymbol{f}_{\text{meas}} - \hat{\boldsymbol{b}}_a) + \boldsymbol{g}(\hat{\boldsymbol{p}}_k) + \boldsymbol{a}_{\text{fr}}(\hat{\boldsymbol{p}}_k, \hat{\boldsymbol{v}}_k), & & \\
\hat{\boldsymbol{p}}_{k+1}^- &= \hat{\boldsymbol{p}}_k^+ + \hat{\boldsymbol{v}}_k^+\Delta t + \tfrac{1}{2}\hat{\boldsymbol{a}}_k\Delta t^2, & \hat{\boldsymbol{v}}_{k+1}^- &= \hat{\boldsymbol{v}}_k^+ + \hat{\boldsymbol{a}}_k\Delta t.
\end{aligned} \tag{24}$$

The environment supplies the gravity and rotating-frame gradients used in the discrete transition matrix $\mathbf{F}_k$. Attitude error couples into velocity through $-\mathbf{C}_{1/2}[\boldsymbol{f}_{\text{meas}} - \hat{\boldsymbol{b}}_a]_\times \delta\boldsymbol{\theta}\Delta t$; gyro bias, scale factors and misalignment couple through the calibrated rate. The symbol $[\cdot]_\times$ denotes the cross-product matrix. Equation (25) propagates uncertainty:

$$\mathbf{P}_{k+1}^- = \mathbf{F}_k\mathbf{P}_k^+\mathbf{F}_k^\mathsf{T} + \mathbf{Q}_k. \tag{25}$$

The process covariance uses the sample-noise PSDs $\sigma_a^2/f_{\text{IMU}}$ and $\sigma_g^2/f_{\text{IMU}}$, plus bias random-walk variance proportional to Δt. For example, the white-acceleration position, position–velocity and velocity blocks scale as $S_a\Delta t^3/3$, $S_a\Delta t^2/2$ and $S_a\Delta t$, respectively. Bias and calibration means remain constant during prediction; enabled random walks grow their covariance.

7.3 Measurement correction and attitude injection

GNSS observes antenna position and velocity. Its lever arm relative to the IMU is removed using the estimated rotation and calibrated angular rate; delayed measurements are extrapolated over their age using the estimated

acceleration. LiDAR uses the analytic ray–sphere range, with position and attitude derivatives through both sensor arm and boresight. Invalid geometry skips correction.

The star-tracker quaternion is converted from ECI to E at its exposure epoch and advanced over its latency using the last estimated rate. Its residual is the shortest rotation vector between the predicted and measured attitude. For each sensor, Equation (26) defines innovation $\boldsymbol{\nu}$, measurement Jacobian $\mathbf{H}$, innovation covariance $\mathbf{S}$, gain $\mathbf{K}$ and error correction:

$$\begin{aligned} \boldsymbol{\nu} &= \boldsymbol{z} - h(\hat{\boldsymbol{x}}^-), & \mathbf{S} &= \mathbf{H}\mathbf{P}^-\mathbf{H}^\mathsf{T} + \mathbf{R}, \\ \mathbf{K} &= \mathbf{P}^-\mathbf{H}^\mathsf{T}\mathbf{S}^{-1}, & \delta\hat{\boldsymbol{x}} &= \mathbf{K}\boldsymbol{\nu}, \\ \widetilde{\mathbf{P}} &= (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}^-(\mathbf{I} - \mathbf{K}\mathbf{H})^\mathsf{T} + \mathbf{K}\mathbf{R}\mathbf{K}^\mathsf{T}. \end{aligned} \tag{26}$$

Additive blocks receive their corresponding error increments. Attitude uses multiplicative injection and covariance reset, Equation (27):

$$\hat{\boldsymbol{q}}^+ = \mathcal{N}[\hat{\boldsymbol{q}}^- \otimes \mathrm{Exp}(\delta\hat{\boldsymbol{\theta}})], \qquad \mathbf{P}^+ = \mathbf{T}_r\widetilde{\mathbf{P}}\mathbf{T}_r^\mathsf{T}, \quad (\mathbf{T}_r)_{\theta\theta} = \mathbf{I} - \tfrac{1}{2}[\delta\hat{\boldsymbol{\theta}}]_\times. \tag{27}$$

Other diagonal reset blocks are identity; the implementation then symmetrizes $\mathbf{P}^+$. The star-tracker gate accepts only innovations with NIS $= \boldsymbol{\nu}^\mathsf{T}\mathbf{S}^{-1}\boldsymbol{\nu} \leq 16.27$, the selected three-degree-of-freedom 99.9% threshold. GNSS and LiDAR NIS values are diagnostic; they do not have equivalent rejection gates.

7.4 Interfaces and interpretation limits

One filter and the same stage-1 sensor stations continue through separation. The upper child has no navigation instance. An optional nine-state supplied-attitude mode provides a diagnostic, while an unaided `GyroAttitude` quaternion provides a dead-reckoning baseline; their results are distinguished in Section 9.

Sensor noise values and initial alignment are assumptions. Nominal true gyro scale factors and misalignments are zero, so estimating these terms does not establish successful nonzero-error calibration. The GNSS generator includes correlated Gauss–Markov position errors, but the filter has no corresponding bias state and uses the white-noise measurement variances. Accelerometer scale error is likewise not an estimated block. These mismatches can affect consistency and require assessment alongside point-estimate accuracy. Section 9.4 reports both errors and uncertainty diagnostics, including accepted star-tracker updates.

8 Mission Execution and Numerical Implementation

The mission layer connects the models of Chapters 1–7. Commands are A-class project designs: official schedule entries anchor selected events and state-triggered conditions determine return transitions. This chapter defines execution; observed times belong in Section 9.

8.1 Ascent and upper-stage commands

The ascent program tilts the nose away from local vertical. Equation (28) defines its target angle and ideal angular-acceleration command, with $x = \max[(t - 8\,\mathrm{s})/(100\,\mathrm{s}), 0]$ and Body attitude error $\boldsymbol{e}_R$:

$$\theta_{\mathrm{cmd}} = 80^\circ \frac{x^2}{1+x^2}, \qquad \boldsymbol{\alpha}_{\mathrm{cmd}} = k_p \boldsymbol{e}_R - k_d \boldsymbol{\omega}, \qquad k_p = 2\,\mathrm{s}^{-2}, \quad k_d = 2.5\,\mathrm{s}^{-1}. \tag{28}$$

Here $\boldsymbol{e}_R = \mathbf{C}_{BE}^{\mathsf{T}}(\boldsymbol{n} \times \boldsymbol{n}_{\mathrm{cmd}})$, with actual and commanded unit nose directions in E. The implemented direction uses projected launch-east. Although the configuration contains a 45° heading field, the current controller does not use that field in constructing the target; no reconstructed ISS ascent plane is claimed. The pitch timescale and operating point were calibrated against the ascent record used again in Section 9.3.

All nine engines initially run at full throttle. The shutdown envelope of Equation (11) reduces both thrust and flow from 140 to 141 s; zero-flow guards then permit separation at 144 s. The conserved upper child ignites at 152 and cuts off at 545 s: throttle 0.95, one-second ramps, 200 km height target and a 0.05 s maximum step. Height/radial-speed feedback selects attitude. Its drag uses $C_D = 0.3$ on the stage cross-section, not the declared `second_stack` configuration.

8.2 Return commands and fuel management

Table 10 collects the principal return settings. The three-engine set is the central engine and an opposite outer pair (indices 0, 1, 5); landing uses only the central engine. Restart guards require separation, zero flow, engines off, available fuel and assumed propulsion readiness, plus the required descent condition and pointing error below 25°. Propellant settling and ignition health are assumed rather than simulated.

Setting	Project value and use
Planning fuel allocations	Boostback 38 t; entry 16 t; landing 10 t; protected reserve 2 t
Boostback	Throttle 1; 10–60 s planning window; predicted-miss target 500 m
Entry	Descending through 45 km; throttle 0.8; shutdown target 500 m s^{-1}
Landing ignition	Reduced stopping predictor, factor 1.05 and 1.5 km clearance allowance
Landing targets	Contact vertical velocity −0.5 m s^{-1}; lateral gains 0.0576 s^{-2} and 0.48 s^{-1}
Pointing limits	Tilt 20°, terminal 4° below 20 m; rate 0.22 rad s^{-1}; acceleration 0.12 rad s^{-2}
Ideal-couple authority	2 MN m during burns; 200 kN m in coast

Table 10 A-class return settings from `data/crs10_mission.json`. Fuel allocations guide burn termination; they are not measured CRS-10 reserves. The limiting-tank residual remains separate from the protected reserve.

Boostback points against the reduced-order predicted impact miss. The predictor advances central gravity, frame acceleration, approximate drag and a prospective entry burn in 0.5 s steps; it is refreshed once per second. The pointing vector is proportional to $-\hat{\boldsymbol{e}}_{\mathrm{miss}}/(8t_f) + 0.8\boldsymbol{u}$, where t_f is predicted flight time and $\boldsymbol{u}$ local up. The first satisfied miss, fuel or duration condition starts shutdown; a 500 kg margin anticipates ramp consumption.

Entry starts on descending altitude and ends on speed or fuel cap. A constrained force solver selects pointing from desired vertical/lateral acceleration and aerodynamic reaction. Landing starts when the 0.1 s stopping predictor leaves no more than the declared clearance allowance.

Landing feedback operates on the lowest leg tip. If h_ℓ and $\dot{h}_\ell$ are its clearance and radial velocity, the commanded contact acceleration is $[-\dot{h}_\ell|\dot{h}_\ell| - v_t^2]/[2\max(h_\ell, 0.005\,\mathrm{m})]$, with target v_t from Table 10. Rotational, moving-CG and ground-curvature corrections convert this to the required CG acceleration. Lateral position and velocity feedback drive the assumed target. Unused boostback/entry allocations may transfer to landing, while the reserve and limiting-tank residual are protected; a 200 kg margin anticipates landing shutdown. Contact geometry and acceptance limits are defined in Section 3.

8.3 Force/moment allocation and actuator representation

The force solver respects the tilt envelope. Throttle ramps are numerical envelopes, not additional validated steady operating points. Fins receive static deflections through Section 5; travel-rate dynamics are absent.

Equation (29) distinguishes the ideal attitude input from physical TVC:

$$\boldsymbol{M}_{\text{ideal}} = \mathbf{J}_C \boldsymbol{\alpha}_{\text{cmd}}, \qquad \mathbf{J}_C^{-1} \boldsymbol{M}_{\text{ideal}} = \boldsymbol{\alpha}_{\text{cmd}}. \tag{29}$$

The ascent couple is unlimited. Return control adds gyroscopic feedforward and aerodynamic-moment compensation, allocates fin demand, then saturates the residual ideal couple at the declared authority. Changing-inertia and exhaust-flow terms still enter the physical dynamics. Inertia cancellation in the commanded contribution is model inversion, not a loss of inertia or proof that aerodynamic stability removes the need for TVC. `Actuators.py` remains empty and the propulsion evaluator rejects nonzero gimbal angles. Thus landing results test an assumed moment authority, not nozzle/servo dynamics or closed-loop robustness to actuator/model error.

8.4 Integration, event and sensor timing

The accepted-state loop follows this order:

1. Evaluate phase conditions and accept a due event; apply engine/configuration changes and preserve both children at separation.
2. Generate due sensor samples and update navigation. Record truth/estimate output when due.
3. Select a step bounded by 0.01 s, scheduled events, control/ramp knots, output time and the next sensor clock. Near contact, clearance and speed impose a tighter bound.
4. Apply RK4 to the physical state with trial-state loads and mass properties, then synchronize the accepted vehicle object.

RHS evaluations are pure with respect to clocks and random sampling. Guidance normally updates every 0.2 s; landing adapts that interval between 0.02 and 0.2 s. Sensors retain their independent clocks from Table 8, and output is normally every 0.1 s. The internal translation representation follows Section 3.

When a trial crosses the ground sphere at a leg tip, bisection refines the contact epoch with up to 35 iterations and a 10^{-9} s interval criterion. Acceptance uses the pre-contact state; position, speed or attitude are not reset to create success. Propellant-limit, model-domain and unmet-event failures produce explicit run status. A full run requires both an acceptable contact and completion of the intended return sequence.

8.5 Two-vehicle execution and reproducible operation

The booster is advanced until contact or failure. The saved upper-child state is then propagated on its own time interval, using the same mission clock; the two trajectories are physically independent after separation even though their computations are executed sequentially. Navigation remains attached to stage 1.

From the repository root, install the pinned dependencies and execute:

```
python -m pip install -r requirements.txt
python main.py -case full -no-show
```

The full case requests 600 s; booster output ends at contact and the upper child continues. `-output` selects a fresh directory; `-plot-only` reopens saved results without integration.

Outputs are truth for both vehicles, measurements, navigation, events/timeline, ascent comparison data and `summary.json`. The summary retains settings, seed, status and input hashes. Section 9 identifies the records behind its figures and tables.

9 Verification, Validation, and Simulation Results

This chapter separates three questions: whether the implemented equations and interfaces pass their checks, how the generated trajectory compares with public evidence, and how well navigation estimates the simulator's own truth. The nominal record R1 is `results/gyro-final-review-20261110`: seed 1, 0.01 s maximum step, 100 Hz IMU and 0.1 s output. Its runtime input hashes match the present physical/navigation code and data. Figures and numerical tables are regenerated from its saved CSV/JSON files; the file hashes and derived metrics are retained in `report/figure_sources.json`. No new full-flight integration is required to reproduce these plots.

9.1 Model verification and state continuity

The vehicle/navigation suite contains 115 passing assertion-based tests; the environment suite adds 41. They cover mass aggregation and derivatives, force-arm signs, fixed-mass free fall and constant-moment rotation, rotating Body storage, propellant bounds, aerodynamic flow reversal/damping, event guards, separation and two-vehicle execution. These tests establish behavior in their fixtures; they do not measure agreement with the real vehicle. The additional gyro/star-tracker script prints finite-difference and calibration diagnostics without asserting their error tolerances, so it is not presented as complete independent certification of the filter Jacobians.

At 144 s, parent mass is 211019.374 kg and child masses are 91089.374 and 119930.000 kg. Their mass residual is zero. Relative linear- and angular-momentum residuals are 9.95×10^{-19} and 9.54×10^{-12}; recombined-CG position error is 4.66×10^{-10} m. These satisfy the numerical criteria of Section 3 for the assumed zero-impulse map. Both child states remain finite and are propagated without replacing their inherited initial conditions. Conservation under an assumed map does not establish the accuracy of real separation impulses or discarded hardware.

9.2 Continuous mission and separated trajectories

Figure 7 shows the complete generated sequence, and Figure 8 distinguishes the two children. Shutdown completes at 141 and separation occurs at 144 s. The three return burns and their measured consumption appear in Table 11; their endpoints are integration results of the conditions in Section 8.

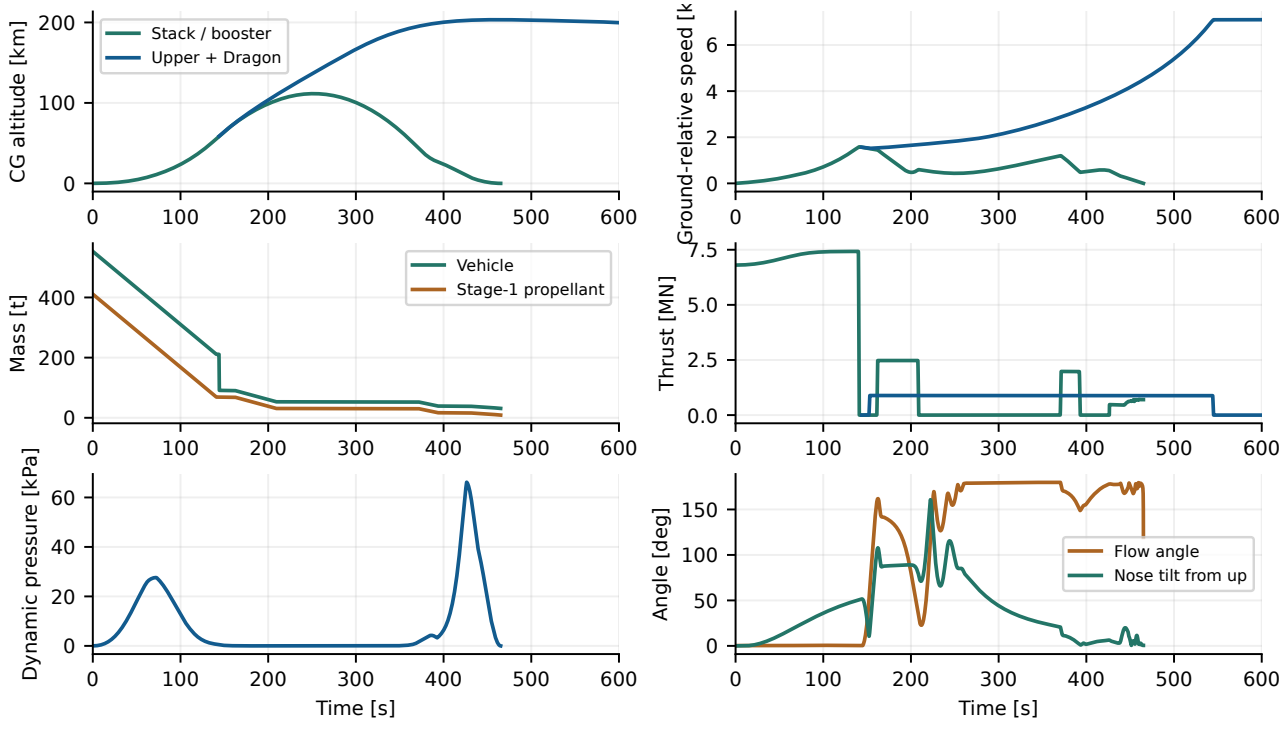


Figure 7 R1 continuous flight. Green denotes stack/booster and blue the upper child where both are shown. The mass jump is a change of tracked configuration at separation. Flow angle near 180° means engine-first relative flow, not attitude-estimation error. The trajectories use project commands and truth-state feedback.

Burn	Start–end [s]	Consumed [t]	End condition
Boostback	161.00–208.81	37.982	Fuel/duration planning cap
Entry	370.14–392.95	14.158	Speed target, then smooth shutdown
Landing	426.16–465.083	7.818	First contact

Table 11 Observed R1 return-burn intervals and fuel consumption. Boostback reaches its planning cap rather than the 500 m predicted-miss target. Its recorded predicted miss at burn end is about 3.97 km; later phases provide additional correction.

The ascent dynamic-pressure maximum is 27.62 kPa at 71.9 s, compared with the approximate 75 s preflight marker. The full-flight maximum is instead 66.16 kPa near landing ignition at 426.3 s; the two maxima must not be confused. Between entry shutdown and landing ignition, gravity accelerates the unpowered descending booster again, explaining the late rise in its speed. Landing then brakes it to the contact state.

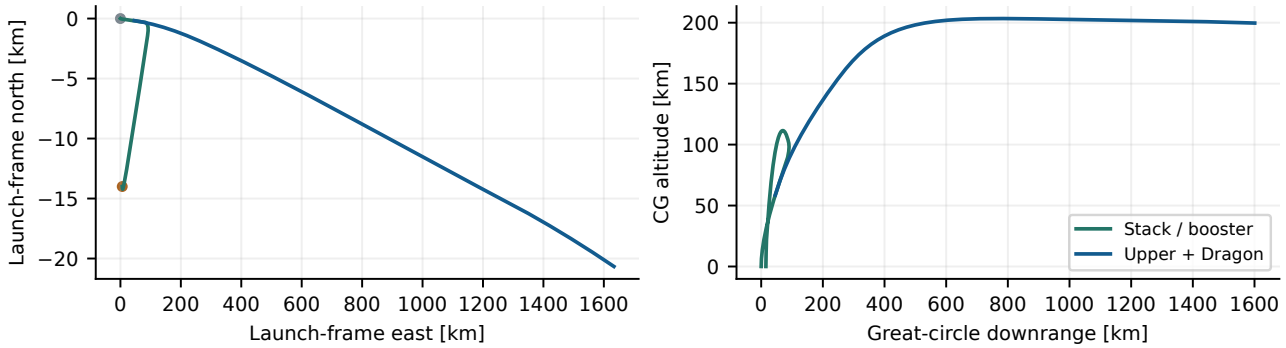


Figure 8 R1 separated paths: launch-frame horizontal projection and altitude against spherical downrange. The marked horizontal points are launch and the assumed return target. The booster returns while the upper-stage/Dragon child continues downrange from the same separation event.

Table 12 evaluates the project limits at contact. Contact occurs at 465.0827 s with 8.931 t of stage-1 propellant remaining. The ground condition applies to the lowest leg tip, so the CG remains above zero altitude. The upper child continues to 600 s, reaching 199.694 km and 7.097 km s^{-1}, with 5.472 t of propellant. This is a successful project scenario, not proof of SpaceX guidance reconstruction or validated orbital insertion.

Acceptance metric	R1 result	Project limit
Vertical speed [m s^{-1}]	0.4975	2
Horizontal speed [m s^{-1}]	0.9016	1
Tilt [deg]	0.5349	5
Angular-rate magnitude [deg s^{-1}]	0.01289	1
Horizontal target error [m]	3.306	10

Table 12 R1 mission-level contact metrics. Speeds and target error use the structural-base state; ground contact is tested at the lowest leg tip. Ground contact and deployed legs are also required and satisfied. These are assumed project limits; horizontal speed has the smallest fractional margin.

9.3 Comparison with the ascent reference

Figure 9 compares 140 retained video-derived samples over 0–139.973 s [13, 14]. Altitude RMSE is 1.577 km and speed RMSE 18.46 m s^{-1}. At the last sample, simulated altitude is 54.628 km versus 59.400 km, a −8.03% difference; speed is 1574.19 versus 1609.44 m s^{-1}, a −2.19% difference.

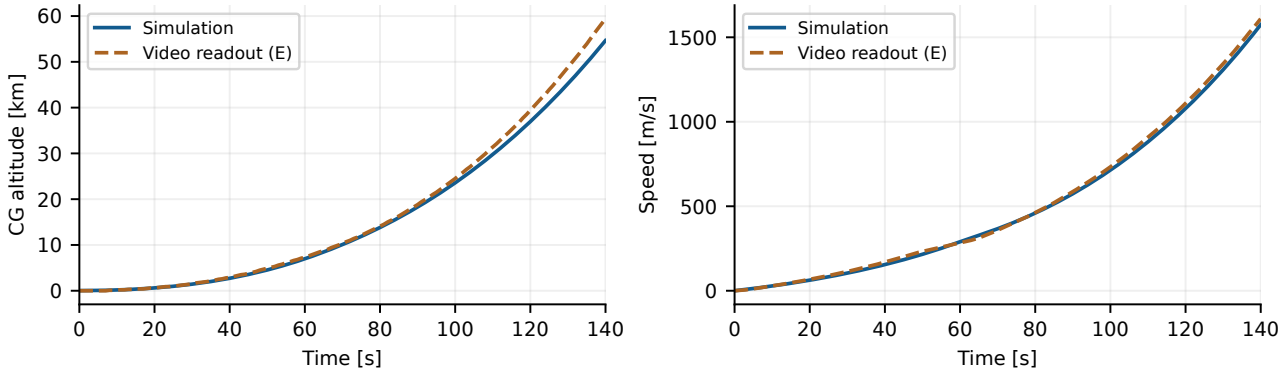


Figure 9 R1 ascent against the E-class digitized broadcast readouts. This is the same record used to calibrate the ascent operating point/pitch timescale. Model CG altitude and ground-relative speed are compared with readouts whose detailed datum/frame conventions are not independently established.

The comparison supports quantitative calibration within the retained ascent interval. It does not independently validate the tuned parameters, establish internal mass properties, or validate the return/upper-stage commands. Readout rounding, digitization and speed-frame uncertainty also limit interpretation of the residuals as pure physical-model error. The fairing-to-Dragon aerodynamic transfer and mixed-source vehicle data remain model uncertainties.

9.4 Navigation accuracy and statistical diagnostics

The gyro-driven navigation result is evaluated at the fixed IMU station, not at the moving CG. Equation (30) defines the three-dimensional position RMS and shortest quaternion-error angle used below:

$$\mathrm{RMS}_p = \sqrt{\frac{1}{N}\sum_{k=1}^{N} \|\hat{\boldsymbol{p}}_k - \boldsymbol{p}_{\mathrm{true},k}\|^2}, \qquad e_q = 2\arccos\left(\left|\hat{\boldsymbol{q}}^{\mathsf{T}}\boldsymbol{q}_{\mathrm{true}}\right|\right). \tag{30}$$

The quaternion inner product is clamped numerically to $[0, 1]$. Velocity RMS uses the corresponding velocity error. Figure 10 shows accuracy and uncertainty diagnostics together.

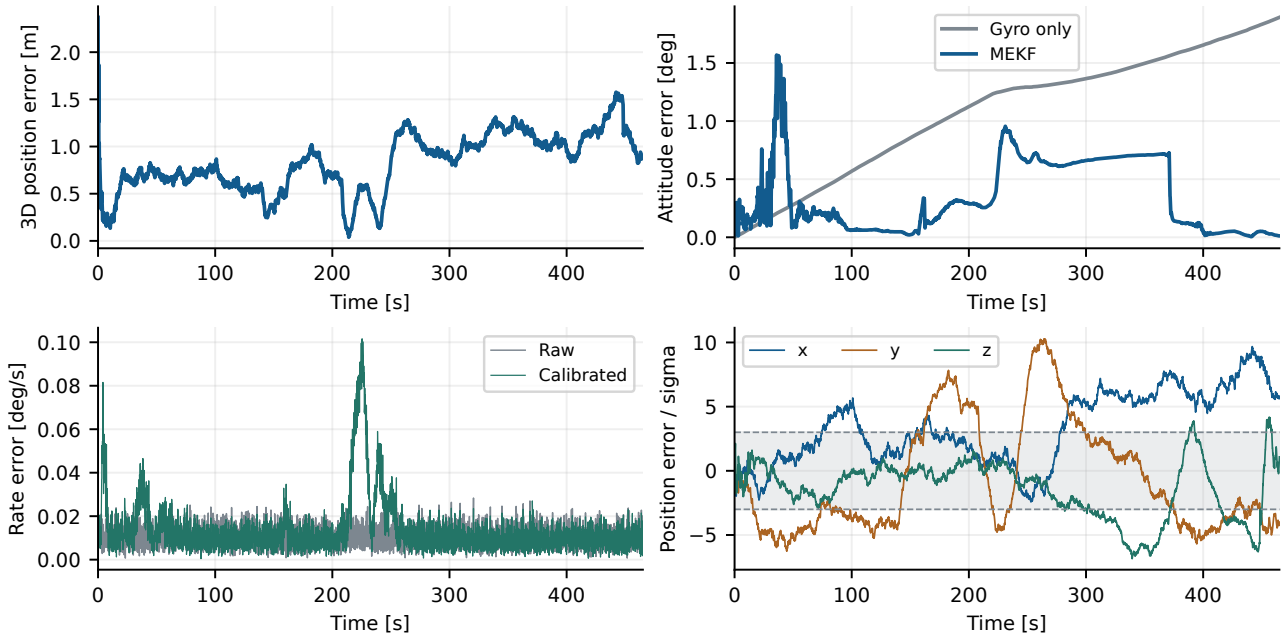


Figure 10 R1 navigation. The MEKF improves the accumulated attitude error relative to the paired unaided gyro quaternion, but this does not imply calibrated covariance or improved instantaneous rate accuracy. Bottom right: Earth-frame position error divided by reported per-axis standard deviation; the shaded interval is ±3.

Position RMS is 0.881 m and velocity RMS 0.1354 m s^{-1}. Attitude RMS is 0.456°, peak 1.568° near 35.9 s, and final error 0.01214°. The paired gyro-only baseline grows overall to 1.892°. These are simulation-truth errors under the declared initial alignment and sensor settings. An earlier supplied-attitude diagnostic is not used to claim a controlled gain in position accuracy because its sensor implementation/realization differs.

The final estimated z-gyro bias is 1.0138×10^{-4} rad s^{-1}, about 1.38% above the configured constant term. This comparison excludes the separately generated random-walk realization. Across 4651 synchronized output/IMU epochs, calibrated angular-rate RMS error is 0.01898° s^{-1}, versus 0.01152° s^{-1} for the raw gyro. Attitude-drift improvement therefore does not establish improved instantaneous angular-rate estimation or successful identification of every calibration parameter.

LiDAR supplies 496 valid updates. All 381 valid star-tracker exposures are rejected by the NIS gate, so R1 has no accepted star-tracker aiding. The mean GNSS NIS is 7.58 for six innovation components; that mean alone does not demonstrate statistical consistency. The measured time fractions within reported position $\pm3\sigma$ are 50.0%, 34.4% and 75.2% by axis. These correlated single-run diagnostics show that the uncertainty envelopes often understate observed error; they are not Monte Carlo coverage probabilities, and full-covariance NEES is unavailable from the logged diagonals.

Plausible contributors include correlated GNSS errors without an estimated bias block, unestimated accelerometer scale error, nonlinear maneuvers and the calibration-state linearization. The available record does not isolate one cause of all rejected star fixes. Covariance modeling and initial acquisition/gating need further study before claiming consistent uncertainty or effective absolute-attitude aiding.

9.5 Numerical sensitivity and demonstrated scope

A saved variant, `results/two-stage-final-check`, uses 0.05 s steps and 20 Hz IMU with matching compared core-physics input hashes. It also lands, at 465.1900 s, with target error 6.171 m and tilt 1.806°, versus R1's 3.306 m and 0.535°. The plan differs in IMU rate and the integration step is also changed. This is a joint step/clock diagnostic, not an isolated RK4 convergence-order study; contact metrics remain numerically sensitive even when both cases pass.

A separate static allocation check quantifies the dry-CG uncertainty of Section 3. With an assumed 10 t of propellant remaining, the booster mass is 32.2 t and nominal CG station 16.211 m. Moving the unallocated 6450 kg centroid over the declared 4.0–40.8 m interval shifts the combined CG by up to ±3.686 m. For an explicitly assumed 1 kN transverse force at a fixed structural station, this changes its CG moment by up to ±3.686 kN m. These are conditional allocation effects, not flight-error confidence bounds.

R1 demonstrates finite continuous ascent, conservative separation, two-child propagation, project-designed booster contact, and gyro-driven navigation with quantified errors. It does not establish return robustness to wind/coefficient uncertainty, nonzero calibration errors or navigation-fed control. Physical TVC dynamics, restart transients and independent return-flight validation remain outside the demonstrated evidence. These limits define what later GNC design may safely infer from the present simulator.

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